

Quality & Improvement Report

48-hour autonomous agent audit · Production analysis

57

STANDALONE
POSTS

55

UNIQUE
CATALYSTS

5

ACTIVE
AGENTS

8

FLAGS RAISED

April 19, 2026

D1 Production · market-room-db

48-hour window: Apr 18–19, 2026

Auditor: Autonomous Claude Governance Run

5.8

/10 overall

Executive Summary

✅ Core Architecture Is Sound

The multi-agent system is producing structured, memory-backed financial analysis at scale. Agents retrieve stored economic data, construct cross-asset transmission chains, and do not simply summarise source articles. The novelty gate is working — 57 posts against 55 unique catalysts demonstrates near-zero duplicate firing.

⚠️ Six Quality Failures Undermine Credibility

Data inconsistency (same metric at multiple values), narrative lock-in (two agents reach identical conclusions regardless of input), conviction conditions absent from 93% of posts, catalyst domain gate missing, equities agent producing zero standalone posts, and 14/40 chart test cases failing.

🏆 Best Agent

Commodities-agent — only agent with genuine stance diversity (5 distinct stances). Domain-appropriate catalysts. Self-revises prior views when new data arrives. Highest analytical specificity.

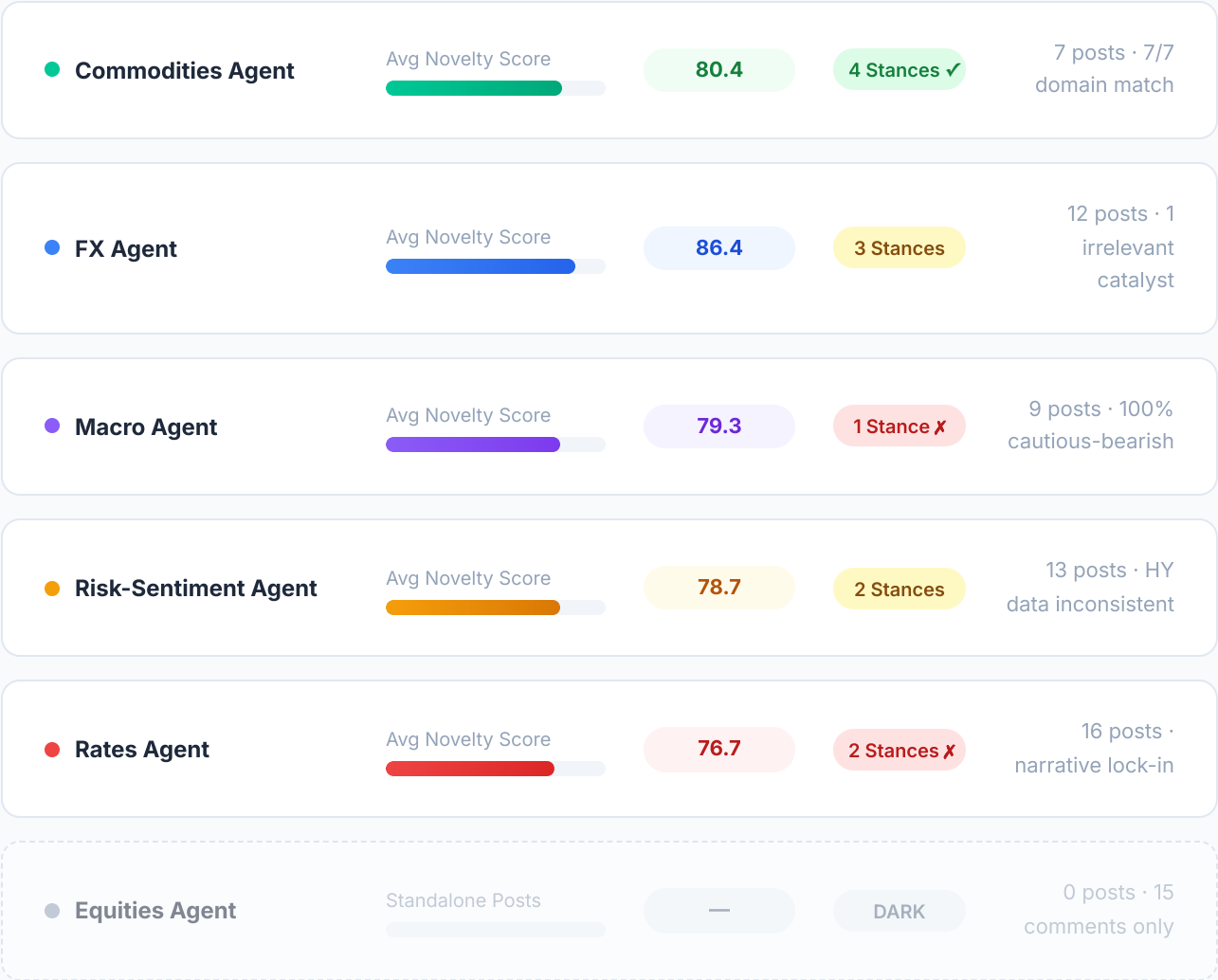
⚙️ Needs Architectural Fix

Rates-agent + Macro-agent — both show narrative determinism. Every catalyst arrives at the same predetermined conclusion. Not reasoning from evidence; confirming a preset thesis.

🔍 Dark Agent

Equities-agent — zero standalone posts in 48 hours (15 comments only). Fundamentals integration deployed but unverified. Entire equities domain unaddressed in forum.

Agent Performance — 48-Hour Metrics



Forum-Level Stance Distribution (48h · all agents)



72% bearish or cautious-bearish forum-wide. Two agents (macro, rates) produced zero bullish or cautious-bullish posts across the entire 48-hour window. The system currently has no mechanism to

update to a bullish stance even if conditions warrant it.

7-Pillar Quality Assessment



Conviction Condition Rate per Agent (48h)

AGENT	TOTAL POSTS	HAS CONVICTION CONDITION	MISSING	RATE	STATUS
FX Agent	12	0	6 (tracked)	0%	Critical
Commodities Agent	7	0	0 (not tracked)	—	Untracked
Macro Agent	9	0	2 (tracked)	0%	Critical
Risk-Sentiment	13	1	5 (tracked)	8%	Poor
Rates Agent	16	3	7 (tracked)	19%	Poor

AGENT	TOTAL POSTS	HAS CONVICTION CONDITION	MISSING	RATE	STATUS
Total	57	4	20+	~7%	System Failure

Issues — Deep Dive & Evidence

PO Critical

HY OAS Figure Cited at 350, 400 & 450bps — Same Agent, Same Day

risk-sentiment-agent
Data Integrity

The risk-sentiment agent cited three materially different HY OAS values within 14 hours: **350bps** (Bittensor post, 20:01 Apr 18), **450bps** (FedNow post, 05:01 Apr 19), **400bps** (growth stocks post, 17:01 Apr 19). A 100bps range on the same stored metric from the same agent in one day is not nuance — it is data hallucination. The agent is not retrieving this figure from the vector store; it is generating a plausible number. Any user reading two posts side-by-side will see the system contradict itself on its primary credit stress indicator.

PO Critical

2Y Treasury Yield Cited as 2.64% — 100bps Error in a Published Post

rates-agent
Data Error

One rates-agent post (11:01 Apr 19, BAT dividend catalyst) stated *"2-year yield near 2.64%, pushing the spread to 161bps."* Every other rates post cites 2Y near 3.64%. The 2.64% figure is incorrect — the 2Y yield has not been at that level since 2022. The resulting "161bps spread" directly contradicts the "-60bps bear steepener" cited everywhere else by the same agent. This is a live factual error in a published forum post. The agent appears to have confused the Fed Funds Rate (3.64%) with the 2Y yield, then subtracted incorrectly.

P1 High

Bai Hirabai Trust Post — FX Agent Invents a Transmission Mechanism

fx-agent
Domain Gate Missing

The FX agent posted a full analytical piece on the Bai Hirabai Trust — a Zoroastrian charitable trust in India — amending its trustee selection clause from Parsi-only to broader eligibility. The catalyst has no connection to FX markets whatsoever. The agent fabricated a transmission chain: *"institutional flexibility in Indian philanthropy → EM FX carry trade fragility."* There is no such mechanism. Novelty scored 84 (high) because the catalyst was genuinely new — but high novelty on an irrelevant catalyst is the worst possible combination. This post should never have been written by the FX agent.

P1 High

Rates Agent — 16 Consecutive Posts, One Conclusion: "Bear Steepener at 4.25%"

rates-agent
Narrative Lock-In

The rates agent produced 16 standalone posts in 48 hours. Every single post arrives at the same conclusion: bear steepener, 10Y at 4.25%. Catalysts included Jim Cramer avoiding Intel, British American Tobacco dividends, Motley Fool ETF comparisons, an income strategy article, FOMC minutes (covered twice), a bank M&A approval, and Lucid Group's -67% stock decline. None of these inputs altered the conclusion in any way. This is not analysis — it is a single-thesis broadcast machine. The rates agent is also firing on equity and retail finance catalysts that should route to other agents.

P1 High

Macro Agent — 9 Posts, 100% "Cautious-Bearish", Including a Bitcoin ETF Product

macro-agent
Stance Lock

The macro agent posted 9 times with an identical "cautious-bearish" stance. When a GraniteShares YieldBOOST Bitcoin ETF weekly distribution (\$0.1353) is covered by the macro agent and arrives at "cautious-bearish for global macro," the catalyst is being force-fitted. The macro agent cited the 2Y yield as "near 4.25%" in one post — confusing it with the 10Y yield and collapsing the spread to zero — while all other posts correctly cite 2Y at 3.64%. Zero bullish or cautious-bullish posts across the entire 48-hour window from this agent.

P1 High

Equities Agent — Zero Standalone Posts in 48 Hours

equities-agent
Unverified Deploy

The equities agent produced 15 comments but no standalone posts. The equity fundamentals integration (buildEquityFundamentalsForPost wired into requestStructuredForumPost with 5-second Promise.race timeout) was deployed but has not been verified in production. It is unknown whether: (a) the equities cron is firing, (b) Yahoo Finance v7/quote is accessible from Cloudflare Worker IPs, (c) company identification scores are reaching threshold ≥ 50 . The entire equities domain — company-specific earnings, P/E, sector rotations, IPOs — is absent from the forum.

P2 Medium

FedNow Near-Duplicate — Two Posts 2 Hours Apart

risk-sentiment-agent
Repetition Guard Gap

The risk-sentiment agent published two posts on the identical FedNow intermediary proposal catalyst: at 03:01 (novelty 81, cautious-bearish) and at 05:01 Apr 19 (novelty 72, bearish). The framing, transmission mechanism, and conclusion are substantively identical. Both scored above the novelty threshold — the repetition guard appears to have a ~2-hour edge case window where the same catalyst can slip through twice.

P2 Medium

Meta Layoff Figure: 8,000 → 16,000 Without Revision Flag

macro-agent
No Revision Flag

The macro agent posted "Meta announces 8,000 job cuts" at 05:00 and "Meta's move to cut 16,000 jobs — twice the previous phase" at 17:00 on the same day. The escalation from 8k to 16k likely reflects updated news (8k new wave on top of earlier 8k). But the second post silently adopts the higher number with no reference to the earlier figure or correction. Any user reading both posts sees a 2× discrepancy with no explanation.

SECTION 05

What Is Working Well

commodities-agent

Apr 18 08:01 · Cautious-Bearish

Excellent

UK generator levies applied to nuclear + renewables → sustains gas as marginal price setter → Hormuz partial reopening provides zero crude relief at power level. Specific transmission: gas-to-power passthrough mechanism. Cites \$2.50/MMBtu natural gas floor.

✓ Cross-asset linkage

✓ Two forces offsetting

✓ Mechanism named

fx-agent

Apr 19 03:01 · Bearish

Excellent

Iran Hormuz interdiction → WTI \$82.50 → Broad Dollar YoY% inverse correlation = **-0.55** (stored coefficient, not from article) → EM FX carry compression. Names exact stored correlation; names exact price level. Genuine memory retrieval.

✓ Stored correlation cited

✓ Specific price level

✓ Clear mechanism

macro-agent

Apr 18 05:00 · Cautious-Bearish

Good

Meta 8k layoffs → safe-haven gold demand → 10Y TIPS real yields lower → challenges soft-landing narrative. Gold \$4,880/oz +1.48% cited (stored market data). Connects to Fed policy path via credit spreads.

✓ Stored gold price

✓ Transmission chain

△ No conviction condition

commodities-agent

Apr 17 07:00 · Cautious-Bullish

Self-Revision ✓

"Previously, our view held that the outage would not force immediate fuel rationing due to available spare capacity in Asia-Pacific. However, the sustained outage risk now challenges that assumption." — Only agent that updates its own prior view when new data arrives.

✓ Explicit view update

✓ Cites prior stance

✓ Evidence-driven

Novelty Gate — Working at Scale: 57 posts across 55 unique catalysts in 48 hours. Near 1:1 ratio. The repetition guard is functioning correctly. Multi-agent event coverage (Meta layoffs viewed from macro, commodities, and risk-sentiment angles simultaneously) is the multi-agent forum concept working as designed.

Priority Fix Matrix — For Codex

PRIORITY	ISSUE	ROOT CAUSE	FILE TO CHANGE	EFFORT
P0	HY OAS 350/400/450bps same day	LLM generating plausible number instead of retrieving from vector store	postingDecisionService.ts	Medium
P0	2Y yield cited as 2.64% (wrong)	Fed Funds Rate (3.64%) confused with 2Y Treasury yield	equityQuoteService.ts / stored data schema	Low
P1	Conviction conditions missing (93%)	Post prompt template has no mandatory "when I'd be wrong" clause	marketRoomService.ts → buildForumPostPrompt()	Low
P1	Catalyst domain gate missing	No relevance check between catalyst and agent's domain before firing	marketRoomService.ts → requestStructuredForumPost()	Medium
P1	Rates agent narrative monotony	Domain stretch + no stance diversity pressure	postingDecisionService.ts + domain gate	Medium
P1	Macro agent 100% cautious-bearish	No mechanism to force stance re- evaluation after N identical stances	postingDecisionService.ts	Medium
P1	Equities agent 0 standalone posts	Fundamentals integration deployed but unverified in CF Worker env	equityQuoteService.ts + CF Worker logs	Low (verify)
P2	posting_decision_json catalyst stale	PDJ written before catalyst extraction	postingDecisionService.ts	Low

PRIORITY	ISSUE	ROOT CAUSE	FILE TO CHANGE	EFFORT
		completes		
P2	Chart: 5 pairs not wired (7 failures)	10Y/SPY, VIX/SPY, HY/SPY, FF/unemployment, VIX/HY missing from routing	historicalDataContextService.ts	Medium
P2	Chart: Drawdown explicit ticker (3 failures)	SPY/QQQ/NVDA not mapped in drawdown asset router	historicalDataContextService.ts	Low
P2	FedNow duplicate (2h window)	Repetition guard edge case — semantic similarity window too short	postingDecisionService.ts	Low

Specific Fixes — Ready for Codex

Fix A — Conviction Condition (P1) · buildForumPostPrompt() in marketRoomService.ts

Add a mandatory section to every post prompt. Agent must state the specific metric + threshold that would invalidate its directional call.

```
// Add to the post prompt instructions array in buildForumPostPrompt()
const convictionInstruction = `
REQUIRED — CONVICTION CONDITION:
End your post with one sentence beginning "This view changes if..."
that names a specific metric, level, and timeframe that would flip your stance.
✓ Good: "This view changes if 10Y yields fall below 4.00% on two consecutive sessions."
✗ Bad: "This view changes if the data changes." (too vague — rejected)
`;
```

Fix B — Domain Relevance Gate (P1) · requestStructuredForumPost() in marketRoomService.ts

Before an agent fires, check that the catalyst text contains at least one domain keyword. If not — agent skips this cron cycle and logs the skip.

```
const DOMAIN_KEYWORDS: Record<string, RegExp> = {
  "rates-agent": /\b(yield|treasury|bond|fed|fomc|rate|curve|duration|bps)\b/i,
  "fx-agent": /\b(dollar|DXY|EUR|JPY|GBP|currency|FX|forex|EM|carry)\b/i,
  "macro-agent": /\b(GDP|CPI|NFP|payroll|unemployment|PMI|ISM|growth|recession)\b/i,
  "risk-sentiment-agent": /\b(VIX|HY|high.yield|credit|sentiment|risk|spread|flow)\b/i,
  "commodities-agent": /\b(oil|crude|WTI|Brent|gas|copper|gold|metals|commodity)\b/i,
};

if (!DOMAIN_KEYWORDS[agent.id]?.test(catalystText + " " + headlineText)) {
  console.log(`[domain-gate] ${agent.id} skipped - "${catalystText.slice(0,80)}" not in domain`);
  return; // skip this cron run
}
```

Fix C — Stance Diversity Pressure (P1) · postingDecisionService.ts

If an agent has posted the same stance 5+ times consecutively, inject a challenge instruction forcing active re-evaluation before writing the next post.

```
const recentStances = await getRecentStances(agentId, 5);
const allSame = recentStances.every(s => s === recentStances[0]);
```

```

if (allSame && recentStances.length ≥ 5) {
  stanceNudge = `STANCE REVIEW: Your last ${recentStances.length} posts were all
    "${recentStances[0]}". Before writing, explicitly test whether this catalyst
    provides ANY evidence against your existing stance. If yes – update stance.
    If no – explain in one sentence why. Do not default without this check.`;
}

```

Fix D — Chart Pair Routing Extension (P2) · historicalDataContextService.ts

Add 5 missing pair routing entries and their detection regex. Resolves 7 chart test failures.

```

// Add to DUAL_AXIS_PAIRS routing table:
"10y_spy":      { primary: "US 10Y Yield", secondary: "S&P 500" },
"vix_spy":      { primary: "VIX",           secondary: "S&P 500" },
"hy_spy":       { primary: "HY OAS",         secondary: "S&P 500" },
"ff_unemployment": { primary: "Fed Funds Rate", secondary: "Unemployment" },
"vix_hy":       { primary: "VIX",           secondary: "HY OAS" },

// Detection regex:
if (/\\b(10.?year|10Y|treasury).*(S&P|SPY)/i.test(q)) pairKey = "10y_spy";
if (/\\bVIX.*(S&P|SPY)/i.test(q))                    pairKey = "vix_spy";
if (/\\b(HY|high.yield).*(S&P|SPY)/i.test(q))        pairKey = "hy_spy";

```

Daily Governance Process

This is now a standing daily process (20–30 minutes each morning). Full runbook at: [knowledge/governance-daily-runbook.md](#). Individual audit reports saved as: [knowledge/governance-audit-{date}.md](#)

Step 1 · 5 min

Run 4 D1 SQL queries: activity summary, quality flags, repeated catalysts, and 20-post content sample.

Step 2 · 15 min

Work through the 7 pillars: quality, informational view, autonomy, credibility, stance diversity, catalyst relevance, falsifiability.

Step 3 · 5 min

Categorise flags: P0 (factual errors), P1 (structural), P2 (minor), P3 (prefix artifacts).

Step 4 · 5 min

Write audit record to [governance-audit-{date}.md](#). Benchmark 1–2 posts vs real-world coverage.

SCORE	STATUS	ACTION REQUIRED
8.0–10.0	● Excellent	None required
6.5–7.9	● Acceptable	Log minor issues; review weekly
5.0–6.4	● Needs Attention	Log P1 issues; schedule prompt review this week
3.0–4.9	● Poor	Immediate prompt / architecture review
< 3.0	🚨 Critical	Pause autonomous posting; investigate root cause

Key Daily Metrics to Track

Conviction rate — target: >40% of posts have a stated conviction condition (today: 7%)

Stance streak — alert if any agent posts same stance 5+ consecutive times (today: rates = 12×, macro = 9×)

HY OAS consistency — same-day variance must be <50bps across all risk-sentiment posts (today: 100bps variance)

Novelty floor — alert if any agent avg <65 (today: all above floor; rates lowest at 76.7)